

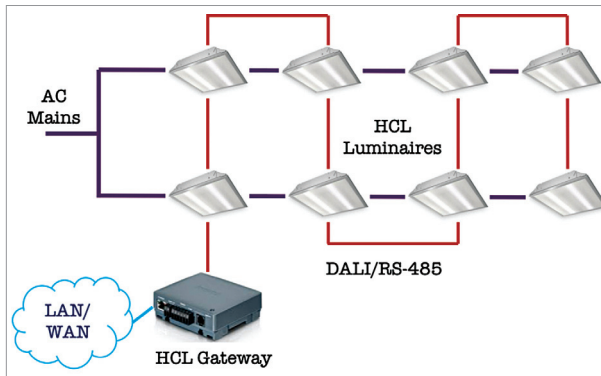


Fig. 6: Topology of a wired human-centric lighting system

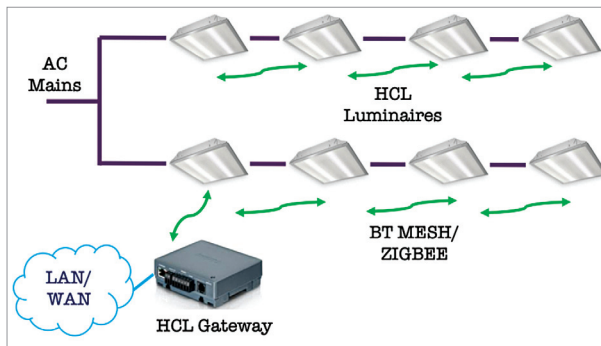


Fig. 7: A classic human-centric/circadian rhythm system

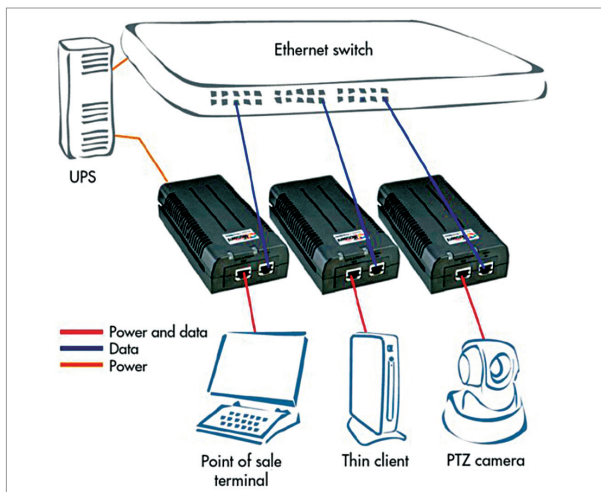


Fig. 8: A classic PoE connection

2. Centralised control system for controlling the human-centric/circadian rhythm lighting systems

The systems can be implemented in three ways:

1. Wired control using RS-485 or DALI protocols

2. Wireless control where every luminaire has an RF interface (ZigBee/6LOWPAN/Bluetooth mesh) and can be controlled through that

3. Power over Ethernet (PoE)

powered luminaires, whose biggest advantage is that lighting system does not need separate main supply as the luminaire can be powered through the Ethernet cable. Also, the system becomes a part of the network infrastructure and is thus easy to manage and control.

Now let us look briefly at each of the controls for better understanding.

Wired system.

Traditional human-centric/circadian rhythm lighting system can be easily controlled through wired connections with the use of wire-line protocols like RS-485 or DALI (a protocol used for controlling stage lighting used in such lighting systems) protocols. This wired control is easy to implement as the existing lighting system can be upgraded to human-centric lighting with just the addition of control wiring and replacement of ordinary luminaires with human-centric luminaires. The system still needs zonal control gateways to integrate into the existing infrastructure. Fig. 6

shows the topology of a wired system.

Wireless system. In the case of wireless control, the performance depends on the building environment in which the human-centric/circadian rhythm lights are used. In most office buildings one of the biggest challenges is that the lights are mounted on a false ceiling. While wired solutions can be implemented through the space between the false ceiling and the roof, wireless systems tend to have per-

formance issues when used in these situations. Also, the wireless band is generic open ISM (industrial, scientific, medical) band and tends to be noisy. It gets disturbed by the environment, especially in industrial environments.

However, when a building is upgraded for human-centric/circadian rhythm system, the above two control systems work well as they are easy to implement. Fig. 7 shows a classic wireless human-centric/circadian rhythm system.

Conventional wireline protocols based systems require laying of new cables for controlling and in some cases the power lines too. At the same time, a wireless system does have less of installation issues but its performance is unpredictable. For a flawless system a reliable solution is needed.

Situations like this can be addressed through Power over Ethernet (PoE) standard. PoE uses the normally available twisted-pair Ethernet cables, which are readily available. It has the ability to control the lights along with supplying power through the same cable, which makes the system easy to implement, use, and maintain.

Introduction to PoE. Today's Ethernet standard allows power to be carried to the end node in addition to the data at speeds ranging from 10Mbps to 1Gbps. A classic PoE connection looks as shown in Fig. 8.

The configuration consists of PoE Ethernet switch, which has a PoE source node (PSE) that sources the power to the end node (PDE) (small black boxes). But as time progresses, most end nodes by default support built-in PoE-PDE support. This means only PoE powered Ethernet needs to be terminated in the equipment (no small boxes needed), which has the PDE function along with its networking port.

IEEE has defined a set of PoE standards whose major variations are in the delivery of power over the Ethernet cable. We need to remember that PoE standards are applicable only for the twisted-pair cable based version of Ethernet. Table 1 shows the different standards defined by IEEE and their power carrying capability.

Fig. 9 shows the cable pairs used for the power distribution by the different PoE standards.